

Day 18 Plan

Maths: DI, Probability, Calendar & Clock · Reasoning: Word Formation, Logical Order, Decision Making · Science: Biology 5: Health, Diseases & Microbes

Today's schedule (about 5½ hours with breaks)

#	Block	Time	What you do	Where
0	Day 17 recall	15 min	10 questions from Day 17, no notes. Redo yesterday's mistakes.	Page 1-2
1	Maths	75 min	5 simplification questions for speed → formulas & methods (10 min) → 2-3 book questions per pattern, timed → 10 Testbook PYQs on this topic → log mistakes	Section A · book · Testbook
-	Break	10 min	Away from the screen	-
2	Reasoning	45 min	Learn the method (10 min) → 15 timed questions, Testbook PYQs first → log	Section B · Testbook
-	Break	10 min	Away from the screen	-
3	Science	110 min	6 lessons with quick checks (≈55 min) → 20-question practice set → Testbook biology PYQs, mixed (finish biology)	Section C · Testbook
-	Break	10 min	Away from the screen	-
4	General awareness	45 min	20 current-affairs MCQs (20 min) → static GA: Science & tech in the news; reports & rankings — Testbook GA PYQs on it (25 min)	CA PDF · Testbook
5	Close	15 min	Redo today's wrong questions · write 3 rules for tomorrow · send me your scores	Mistake log

Short day? Cut in this order: Testbook science PYQs → reasoning down to 10 questions → static GA down to 10 min. Never cut recall, maths or the science lessons. · Exam pace (average 54 sec a question): GA ~30 sec · Science ~45 · Reasoning ~50 · Maths ~65.

Block 0 — Day 17 recall (answers on the last page)

#	Question	Your answer
1	Oxygen in photosynthesis comes from?	
2	Hormone that ripens fruit?	
3	Monohybrid F ₂ phenotypic ratio?	
4	Wings of a bird and an insect are?	
5	Energy passed to the next trophic level?	
6	Chemicals that deplete ozone?	
7	The potato is a modified?	
8	Down syndrome is caused by?	
9	$\tan 60^\circ = ?$	
10	$4 \times 4 \times 4$ painted cube: small cubes with 3 faces painted?	

Fewer than 8 right? Re-read only the lessons you missed — not the whole chapter.

A · MATHS — DI, Probability, Calendar & Clock

Core idea

Three quick-win topics. **DI**: read the units and the question before the chart, then calculate only what is asked. **Probability** = favourable ÷ total outcomes. **Clock angle** has one formula; **calendar** is counting 'odd days' (leftover days after full weeks).

Formulas

Concept	Formula / rule
Percentage share · growth	$\text{part} \div \text{total} \times 100 \cdot (\text{new} - \text{old}) \div \text{old} \times 100$
Pie chart	$\text{Sector angle} = \text{value} \div \text{total} \times 360^\circ \cdot \text{value} = \text{angle} \div 360 \times \text{total}$
Probability	$P = \text{favourable} \div \text{total} \cdot 0 \leq P \leq 1 \cdot P(\text{not } E) = 1 - P(E)$
Sample spaces	1 coin 2 · 2 coins 4 · 1 die 6 · 2 dice 36 · cards 52 (4 suits × 13; 26 red; 12 face cards; 4 aces)
Clock angle	$\theta = 30H - 5.5M $ (take $360 - \theta$ if over 180)
Clock facts (12 h)	Hands coincide 11 times · at right angles 22 times · opposite 11 times (double for a day)
Odd days	Ordinary year 1 · leap year 2 · 100 years 5 · 200 years 3 · 300 years 1 · 400 years 0
Leap year	Divisible by 4; a century year only if divisible by 400 (2000 yes, 2100 no)

Patterns — the method and one solved example each

Pattern	Method	Solved example
1. Pie chart	$\text{angle}/360 \times \text{total}$	Food is 90° of a ₹40,000 budget → ₹10,000
2. Growth	$(\text{new} - \text{old})/\text{old}$	Sales 400 → 500 → 25%
3. One die	favourable/6	Even number → $3/6 = \frac{1}{2}$
4. Two coins	Complement	At least one head → $1 - \frac{1}{4} = \frac{3}{4}$
5. Two dice	Count pairs	Sum 7 → $6/36 = \frac{1}{6}$
6. Cards	Count	A king → $4/52 = \frac{1}{13}$ · a red face card → $6/52 = \frac{3}{26}$
7. Bag	favourable/total	3 red, 5 blue → $P(\text{red}) = \frac{3}{8}$
8. Clock angle	$ 30H - 5.5M $	3:40 → $ 90 - 220 = \mathbf{130^\circ}$
9. Clock angle	$ 30H - 5.5M $	9:30 → $ 270 - 165 = \mathbf{105^\circ}$
10. Next year	Odd days	1 Jan 2026 was a Thursday; 2026 is not a leap year → 1 Jan 2027 is a Friday
11. Days ahead	Remainder ÷ 7	Monday + 61 days → $61 = 8 \times 7 + 5$ → Saturday

Traps

- 'At least one' → use $1 - P(\text{none})$.
- 1900 and 2100 are **not** leap years; 2000 is.
- DI: check whether values are in thousands, lakhs or percentages before calculating.

B · REASONING — Word Formation, Logical Order, Decision Making

Word formation: to check whether a word can be made from a given word, tick off each letter — and check **how many times** it occurs. **Logical order:** arrange by life cycle, size, process or dictionary order. **Decision making:** apply every condition to the candidate one by one; a single failed condition decides the answer.

Patterns, with solved examples

Pattern	Example	Solution
Word formation	Which word cannot be made from EXAMINATION: NATION, MOTION, MAIN, TAME?	MOTION — needs two O's; EXAMINATION has one
Logical order	Arrange: 1 Infant 2 Old 3 Adult 4 Child 5 Adolescent	1, 4, 5, 3, 2
Logical order	Arrange: Plant, Seed, Fruit, Flower	Seed, Plant, Flower, Fruit
Logical order	Arrange: 1 Cut 2 Wear 3 Measure 4 Stitch	3, 1, 4, 2
Decision	Criteria: graduate with $\geq 55\%$, aged 21–30 on 1 Oct 2026, 2 years' experience. Candidate: B.Sc. 60%, born 5 Mar 1999, 3 years' experience.	Age 27 ✓, marks ✓, experience ✓ → selected

Speed rule

In word formation, the trap is almost always a repeated letter — count before you tick.

C · SCIENCE — Biology 5: Health, Diseases & Microbes

6 lessons — the last new science chapter. Health and disease questions are direct recall: pathogen, vector, test, vaccine. Lesson 6 is a one-page table of biology 'records' and 'fathers' to revise again on Day 22.

Lesson 1 · Health, immunity and vaccines (about 7 min)

WHO defines health as complete **physical, mental and social** well-being. Diseases are **communicable** (spread by germs) or **non-communicable**; **acute** (short) or **chronic** (long-lasting).

Immunity: innate (from birth) and acquired. A **vaccine** contains weakened or killed germs (or their parts) and trains the body to make **antibodies**. **Edward Jenner** made the first vaccine (smallpox, 1796). **Louis Pasteur** made the rabies vaccine and invented pasteurisation.

Antibiotics work against bacteria, not viruses.

Quick check 1 — answer out loud before moving on (answers at the end)

a) Who developed the first vaccine? b) Do antibiotics cure viral infections?

Lesson 2 · Microbes — useful and harmful (about 7 min)

Microbe	Use
Lactobacillus	Sets milk into curd
Yeast (Saccharomyces)	Bread (CO ₂ makes it rise), alcohol — fermentation
Penicillium	Penicillin — first antibiotic (Alexander Fleming, 1928)
Streptomyces	Streptomycin
Rhizobium · Azotobacter	Fix nitrogen in soil

Viruses show life only inside a host cell and can be crystallised. The first virus studied was the **tobacco mosaic virus** (Ivanovsky, 1892). Food poisoning (botulism) is caused by *Clostridium botulinum*.

Quick check 2 — answer out loud before moving on (answers at the end)

a) Which bacterium sets curd? b) Penicillin comes from which fungus, discovered by whom?

Lesson 3 • Diseases by pathogen (about 12 min)

Pathogen	Disease — key fact
Bacteria	Tuberculosis (Mycobacterium; BCG vaccine; Robert Koch; Mantoux test) · cholera (Vibrio; contaminated water; ORS) · typhoid (Salmonella typhi; Widal test) · tetanus · diphtheria and whooping cough (DPT vaccine) · leprosy · plague (rat fleas) · pneumonia
Viruses	Common cold, influenza · measles, mumps, chickenpox · polio (oral polio vaccine; India certified polio-free in 2014) · rabies (dog bite) · hepatitis B (liver; via blood) · dengue and chikungunya (Aedes) · AIDS (HIV attacks helper T-cells; ELISA test) · COVID-19 · smallpox (eradicated 1980)
Protozoa	Malaria (Plasmodium; female Anopheles; quinine from cinchona bark) · amoebic dysentery (Entamoeba) · kala-azar (Leishmania; sandfly) · sleeping sickness (Trypanosoma; tsetse fly)
Fungi	Ringworm, athlete's foot
Worms	Filariasis/elephantiasis (Wuchereria; Culex) · roundworm, tapeworm
Vector	Spreads
Female Anopheles mosquito	Malaria
Aedes mosquito	Dengue, chikungunya, Zika, yellow fever
Culex mosquito	Filariasis, Japanese encephalitis
Housefly	Carries germs of cholera, typhoid
Sandfly · rat flea · tsetse fly	Kala-azar · plague · sleeping sickness

Quick check 3 — answer out loud before moving on (answers at the end)

a) Vector of malaria? b) Test used for typhoid? c) Which mosquito spreads dengue?

Lesson 4 • Non-communicable, occupational and pollution diseases (about 7 min)

Disease	Cause
Diabetes · hypertension · heart disease · cancer	Lifestyle, genes; cancer = uncontrolled cell division
Silicosis	Silica dust (stone quarries, glass)
Asbestosis · black lung	Asbestos fibres · coal dust
Byssinosis	Cotton dust (textile mills)
Minamata disease	Mercury in fish
Itai-itai disease	Cadmium
Blue baby syndrome	Nitrate in drinking water
Fluorosis	Excess fluoride in water
Arsenicosis	Arsenic in groundwater

Quick check 4 — answer out loud before moving on (answers at the end)

a) Minamata disease is caused by? b) Occupational disease of stone-quarry workers?

Lesson 5 • Drugs, alcohol and medical tests (about 5 min)

Item	Fact
Nicotine	In tobacco; addictive; tobacco causes oral and lung cancer
Morphine	From the opium poppy; strong painkiller
Alcohol	Long-term use damages the liver (cirrhosis)
ELISA • Widal • Mantoux	HIV • typhoid • tuberculosis
ECG • EEG • MRI	Heart • brain • body scan using magnets

Quick check 5 — answer out loud before moving on (answers at the end)

Which test is used to detect HIV?

Lesson 6 • Biology records and 'fathers' (revise on Day 22) (about 7 min)

Fact	Answer	Fact	Answer
Largest gland	Liver	Largest endocrine gland	Thyroid
Largest artery	Aorta	Largest vein	Inferior vena cava
Longest bone	Femur	Smallest bone	Stapes
Normal temperature	37 °C / 98.6 °F	Normal BP	120/80 mm Hg
Heart rate	≈ 72 per minute	Blood pH	≈ 7.4
Ribs	12 pairs	Chromosomes	46 (23 pairs)
Father of ...	Name	Father of ...	Name
Biology	Aristotle	Medicine	Hippocrates
Genetics	Gregor Mendel	Taxonomy	Carolus Linnaeus
Immunology	Edward Jenner	Microbiology	Leeuwenhoek / Louis Pasteur
Evolution (natural selection)	Charles Darwin	Green Revolution in India	M. S. Swaminathan

Quick check 6 — answer out loud before moving on (answers at the end)

a) Largest artery in the body? b) Normal body temperature in °F?

Science 1-page summary — revise from this tomorrow

Topic	Must remember
Vaccines	Jenner smallpox (1796) · Pasteur rabies, pasteurisation · antibiotics only against bacteria
Microbes	Lactobacillus curd · yeast bread, alcohol · Penicillium → penicillin (Fleming 1928) · TMV first virus
Bacterial	TB (BCG, Mantoux, Koch) · cholera (ORS) · typhoid (Widal) · tetanus · DPT · plague
Viral	Polio (India polio-free 2014) · rabies · hepatitis B · dengue (Aedes, platelets ↓) · AIDS (HIV, ELISA) · smallpox eradicated 1980
Protozoan	Malaria Plasmodium, female Anopheles, quinine · kala-azar sandfly · sleeping sickness tsetse
Vectors	Anopheles malaria · Aedes dengue, chikungunya, Zika · Culex filariasis, JE
Pollution	Minamata mercury · itai-itai cadmium · blue baby nitrate · fluorosis fluoride · silicosis silica
Records	Liver largest gland · thyroid largest endocrine · aorta largest artery · 37 °C = 98.6 °F

Science practice set — 20 questions · target 6 minutes

#	Question
1	The first vaccine (against smallpox) was developed by: (a) Louis Pasteur (b) Robert Koch (c) Alexander Fleming (d) Edward Jenner
2	Antibiotics are effective against: (a) viruses (b) all germs equally (c) fungi only (d) bacteria
3	Curd is formed by: (a) Lactobacillus (b) yeast (c) Rhizobium (d) Penicillium
4	Penicillin was discovered by: (a) Edward Jenner (b) Alexander Fleming (c) Louis Pasteur (d) Robert Koch
5	Yeast is used in making bread because it: (a) produces oxygen (b) adds protein (c) produces carbon dioxide (d) kills bacteria
6	Tuberculosis is caused by a: (a) virus (b) bacterium (c) protozoan (d) fungus
7	BCG vaccine is given against: (a) polio (b) tuberculosis (c) measles (d) cholera
8	The Widal test is used to diagnose: (a) malaria (b) AIDS (c) tuberculosis (d) typhoid
9	Malaria is caused by: (a) a virus (b) Plasmodium (c) Vibrio (d) Entamoeba
10	Malaria is spread by: (a) the female Anopheles mosquito (b) the Aedes mosquito (c) the Culex mosquito (d) the housefly
11	Dengue is spread by the: (a) Anopheles mosquito (b) Culex mosquito (c) sandfly (d) Aedes mosquito
12	Which is a viral disease? (a) Cholera (b) Rabies (c) Typhoid (d) Tuberculosis
13	HIV mainly attacks the: (a) red blood cells (b) helper T-cells (c) platelets (d) liver cells
14	Kala-azar is spread by the: (a) tsetse fly (b) housefly (c) rat flea (d) sandfly
15	Ringworm is caused by a: (a) fungus (b) worm (c) virus (d) bacterium
16	Minamata disease is caused by: (a) cadmium (b) lead (c) arsenic (d) mercury
17	Blue baby syndrome is caused by excess ___ in drinking water: (a) fluoride (b) nitrate (c) arsenic (d) chloride
18	Silicosis affects workers in: (a) textile mills (b) coal mines only (c) fisheries (d) stone quarries
19	The ELISA test is used to detect: (a) typhoid (b) malaria (c) HIV (d) TB
20	The largest artery in the human body is the: (a) aorta (b) pulmonary artery (c) carotid artery (d) vena cava

Score	Time taken	Wrong Q numbers	Mistake type (concept / formula / calc / careless / recall)
/ 20			

Answers

Science practice set

#	Ans	Why / lesson to re-read
1	d	Lesson 1
2	d	Lesson 1
3	a	Lesson 2
4	b	Lesson 2
5	c	Lesson 2
6	b	Lesson 3
7	b	Lesson 3
8	d	Lesson 3
9	b	Lesson 3
10	a	Lesson 3
11	d	Lesson 3
12	b	Lesson 3
13	b	Lesson 3
14	d	Lesson 3
15	a	Lesson 3 — despite its name
16	d	Lesson 4
17	b	Lesson 4
18	d	Lesson 4
19	c	Lesson 5
20	a	Lesson 6

Quick checks

QC	Answers
1	a) Edward Jenner b) No — antibiotics act on bacteria, not viruses
2	a) Lactobacillus b) Penicillium; Alexander Fleming (1928)
3	a) Female Anopheles mosquito b) Widal test c) Aedes
4	a) Mercury b) Silicosis
5	ELISA
6	a) Aorta b) 98.6 °F

Day 17 recall (Block 0)

#	Answer	#	Answer
1	Water	6	CFCs
2	Ethylene	7	Stem
3	3 : 1	8	Trisomy 21 (extra chromosome 21)
4	Analogous organs	9	$\sqrt{3}$
5	$\approx 10\%$	10	8

Send me tonight

Day 17 recall score /10 · science practice score /20 and time · Maths and Reasoning: attempted, correct, average seconds per question · the question numbers you got wrong.